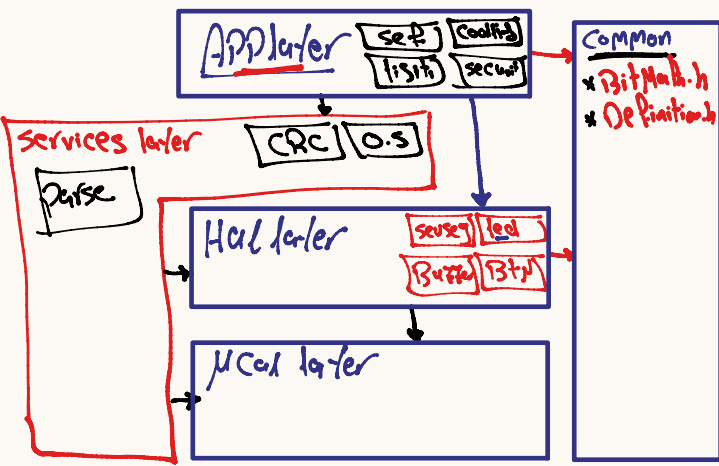


* SW Architecture

↳ layered Architecture



* For each layer Asolate →

* Application layer → write the Algorithm state machine

↳ Colling ↳ security
↳ sound

* HAL layer → Hardware Abstraction layer → write the Module for H.W Component

studio.h
↳ printf

lib → Provide API Function

led_Interface.h
led_on();
led_off();

* MCAL → MC Abstraction layer

↳ write Module for each Peripherals (inside MCU)

* Driver Module

ex → DIO → MCAL
MCAL P
↳ DIO P

↳ DIO_Interface.h
↳ DIO_Private.h
↳ DIO_Config.h
↳ DIO_Program.c

ex: led → HAL

HAL
↳ led

↳ led_Interface.h
↳ led_Config.h
↳ led_Private.h
↳ led_Program.c

Driver → Provide → API's (Function).

① Program.c
↳ Implementation (API's)

② Interface.h
↳ prototype (API's)

③ Private.h
↳ Macros, enums → Drive will be use and can't change
↳ Input → 0
#define Input 0
#define Source 1

④ Config.h
↳ Macro → Can be change.

led driver
#define connection type Sink / Source

* Configuration type

- ① Pre build Config
- ② Post build Config
- ③ Runtime Config

① Pre build Config (Compile-time)

- select Config at compile time
- write Config in file-config.h
- #if ConnectionType == Source
- Fixed → Driver ①

② Post build Config

→ void Direction(A, 1, 1)

A → 1 → output